Quiz 4-B

First and Last Name: ______________________________

You have 10 minutes to complete the quiz. Calculators, notes, and other resources are **not** allowed.

1. (2 points) Let $\mathbf{u} = \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}$. Find the orthogonal projection of $\mathbf{u}$ onto $\mathbf{v}$.

   Show all work.

Solution: The orthogonal projection of $\mathbf{u}$ onto $\mathbf{v}$ is

$$\mathrm{proj}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u}\cdot\mathbf{v}}{\mathbf{v}\cdot\mathbf{v}}\mathbf{v} = \frac{3\cdot(-1)+(-1)\cdot 0 + 2\cdot 2}{(-1)^2+0^2+2^2}\mathbf{v} = \frac{1}{5}\mathbf{v} = \begin{bmatrix} -\frac{1}{5} \\ 0 \\ \frac{2}{5} \end{bmatrix}.$$

QUESTION 2 IS ON THE BACK!!

2. Suppose $\mathbf{w}, \mathbf{x}, \mathbf{y}$, and $\mathbf{z}$ are vectors in $\mathbb{R}^3$, with
$\mathbf{w} \cdot \mathbf{w} = 5$
$\mathbf{w} \cdot \mathbf{x} = 2$
$\mathbf{x} \cdot \mathbf{y} = 0$
$\mathbf{x} \cdot \mathbf{z} = 0$
$\mathbf{w} \cdot \mathbf{z} = 3$
$\mathbf{y} \cdot \mathbf{z} = -2$
Answer each of the following.

(a) (1 point) Find $||3\mathbf{w}||$.

Solution: we have $||\mathbf{w}|| = \sqrt{\mathbf{w} \cdot \mathbf{w}} = \sqrt{5}$. Then $||3\mathbf{w}|| = 3||\mathbf{w}|| = 3\sqrt{5}$.

(b) (1 point) Find $\mathbf{x} \cdot (-15\mathbf{w})$.

Solution: By linearity and symmetry of dot product, we have

$$\mathbf{x} \cdot (-15\mathbf{w}) = -15(\mathbf{x} \cdot \mathbf{w}) = -15(\mathbf{w} \cdot \mathbf{x}) = -30.$$

(c) (1 point) Find $\mathbf{w} \cdot (\mathbf{x} + 4\mathbf{z})$.

Solution: By the linearity of dot product, we have

$$\mathbf{w} \cdot (\mathbf{x} + 5\mathbf{z}) = \mathbf{w} \cdot \mathbf{x} + 4(\mathbf{w} \cdot \mathbf{z}) = 2 + 4 \cdot 3 = 14.$$

(d) (1 point) Is $\mathbf{x}$ orthogonal to $3\mathbf{y} + 4\mathbf{z}$? Fill in the correct bubble below.

○ Yes
○ No
○ Not enough information to answer.

Solution: By the linearity of dot product, we have

$$\mathbf{x} \cdot (3\mathbf{y} + 4\mathbf{z}) = 3(\mathbf{x} \cdot \mathbf{y}) + 4(\mathbf{x} \cdot \mathbf{z}) = 0 + 0 = 0,$$

thus $\mathbf{x}$ is orthogonal to $2\mathbf{y} + 5\mathbf{z}$.